



Replace your bicycle's front wheel with the GeoOrbital wheel to turn it into an electric bike (Courtesy: GeoOrbital)

Instantly turn your bicycle into an electric bike

Convert your bicycle into an electric bike in a few minutes just by replacing the bicycle's conventional front wheel with the GeoOrbital wheel. This wheel is made of an aerospace-grade aluminium uni-body, which houses a brushless DC motor, a Li-Ion battery and a motor controller. It is surrounded by a flat-proof tyre, which is made by filling a conventional tyre with special high-density foam. The kit includes a simple thumb-activated throttle that you can easily clip onto your handlebar.

The GeoOrbital wheel is available in two variants, which vary in size, battery power, weight and range without pedalling. You need to pick the size that fits your bike. The battery is also easily replaceable, so you can always carry a spare along to replace when required.

GeoOrbital works with any speed range, varied transmission components and in-hub brakes, different fork sizes and bike chain technologies. Also, as it does not interfere with your bike's chain or rear wheel, you can get all-wheel drive by using the throttle while pedalling.

So, even after you have installed GeoOrbital, you can still pedal your bike like a normal one, if required. However, since GeoOrbital is heavier and offers more resistance than a normal wheel, the company suggests riding on a lower gear.

The Cambridge-based company sells the product online and delivers worldwide.

Company: GeoOrbital; **Country:** United States of America; **Website:** <http://www.geoo.com/>

INNOVATION

IN EVERY SPHERE OF LIFE

JANANI GOPALAKRISHNAN VIKRAM

A smart case that runs Android on your iPhone

All iPhone users love their iPhone but, unfortunately, not everybody loves the iOS! Those who are used to Android phones often find it very difficult to adapt to the iOS. Chinese company Jijia has developed a smart case called Mesuit that enables you to run an Android-based OS on your iPhone. Just slip on the case, install the associated app and you are ready to go.

The Mesuit dual-SIM case is a fully-functional mobile device powered by MediaTek's octa-core processor. Designed to fit iPhone 6/6S, this case connects to the phone through the Lightning port. It has its own SIM slot, 2GB RAM, 16GB storage capacity and a 1700mAh battery built into it. That means the iPhone user has an additional SIM slot, extra storage to transfer photos and videos from the phone into the case, and seven more hours of battery life.



This case runs an Android-based OS on iPhone 6/6S

Priced at Rs 9990, the Mesuit case is available in India through Yerha.com.

Company: Jijia; **Country:** China; **Website:** <http://www.jijia.com> (in Chinese); <http://www.yerha.com/page/en/mesuit/>

Get active while playing mobile games

Mobile games are fun but these turn us into couch potatoes too. What if you had a mobile gaming system that responded to your physical movements—games that made you jump and jog? That is what inventor Colt Correa hopes to do with DigiBit. Amply funded on Kickstarter, Correa plans to start shipping the first orders to certain countries by November this year.

The DigiBit system consists of small, wearable, sensor-fitted clip-ons that sync with your mobile phone using Bluetooth Low Energy (BLE). You can wear these devices on your hands and feet with the provided bands, or clip them on to your wristband, shoes or socks. For virtual reality (VR) games, there is an additional VR headset.

Fitted with sensors and microcontrollers, these devices sense and calculate your body motion in real time. They beam this information to your mobile device to control game actions.



DigiBit is a body action driven mobile gaming system (Courtesy: DigiBit)

Your body becomes the game controller, and your physical movements move the characters in the game. You can walk, jog, run, jump and dodge. This way you get a good share of fun and exercise!

There are several exciting DigiBit games like Air Hockey and FunnyWings. An open development kit is also available, so you can develop your own games using the Unity 3D game engine.

Creator: Colt Correa; **Country:** United States of America; **Website:** <https://www.kickstarter.com/projects/1039276273/digitbit-is-wearable-active-fun-for-your-mobile-dev?ref=tag>